Diploma in C Programming

Alison Certificate Presentation

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System and Object-Oriented Programming

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Course Overview

Free Online Course

Thorough introduction to C programming language

Key Concepts

Data management, program flow, variables, arrays, pointers, functions

Structure

16 focused modules with assessments

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Modules 1–4: C Basics and First Program

Module 1: Number Systems

Binary, decimal, hexadecimal relevance

Module 2: Programming Basics

Code structure, syntax, directives, execution

Module 3: Program Flow

Functions, routines, return values, logic

Module 4: Your First Program

Write and compile "Hello World"

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Modules 5–8: Data Types and Variables

Module 5: First Assessment

Check understanding of basics

Module 6: Numbers

Integers, floating-point, ASCII

Module 7: Variables

Declare, initialize, manipulate

Module 8: Arrays

Store and access using indexes



Modules 9–12: Strings, Pointers, and Constants

Module 9: Strings

Character sequences and manipulation

Module 10: Pointers

Memory addresses and arithmetic

Module 11: Constants

Using #define and const

Module 12: Combined Use

Pointers, constants, and strings together

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Modules 13–16: Logic and Looping

Module 13: Statements & Logic

If, else, switch decisions

Module 14: Logical Operators

AND, OR, NOT operators

Module 15: GOTO & Loops

Goto statement and while loops

Module 16: Final Assessment

Comprehensive knowledge test

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Key Concepts Covered

Number Systems

Binary and hexadecimal in code

Program Structure

Syntax, functions, arguments

Data Types

Integers, floats, characters

Memory

Pointers and management

Control Flow

Logic statements and loops

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Practical Applications

Build C Programs

Small to medium scale coding skills

Embedded Systems

OS development and hardware interaction

Foundation for Languages

Advance to C++ and Java

Tools Used

GCC compiler, editors, console interface

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Certificate Showcase

Recognized Credential

Validates C programming fundamentals

Resume Booster

Enhances entry-level programming opportunities





Conclusion and Q&A

Skills Equipped

Essential programming foundation

Career Base

Strong start for software developers

Engagement

Questions welcome to deepen understanding

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